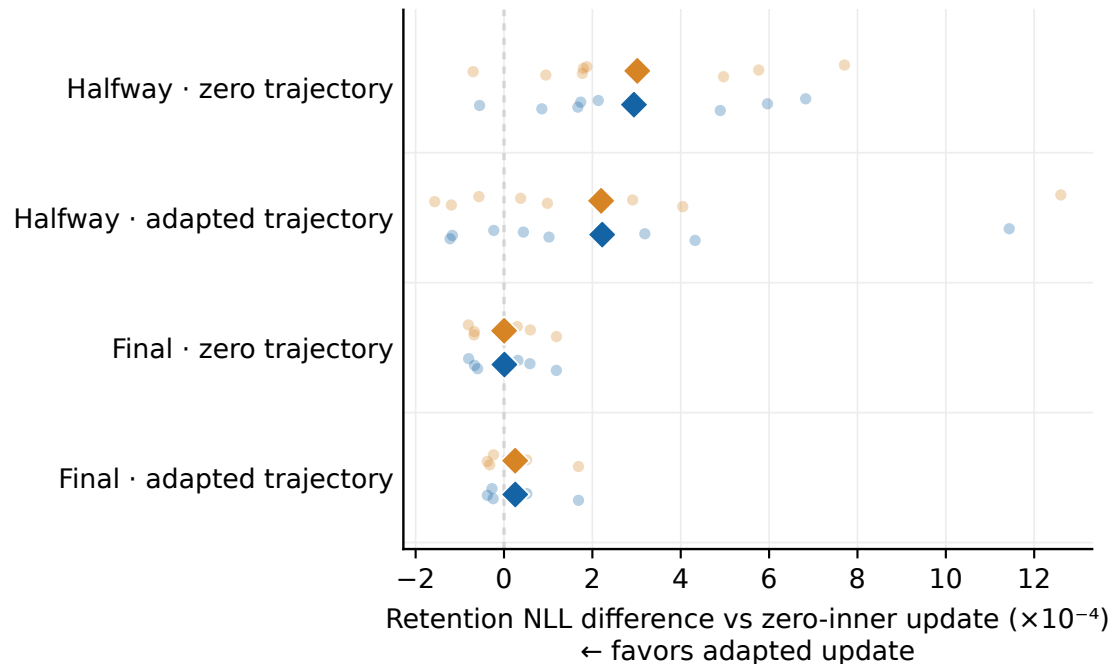


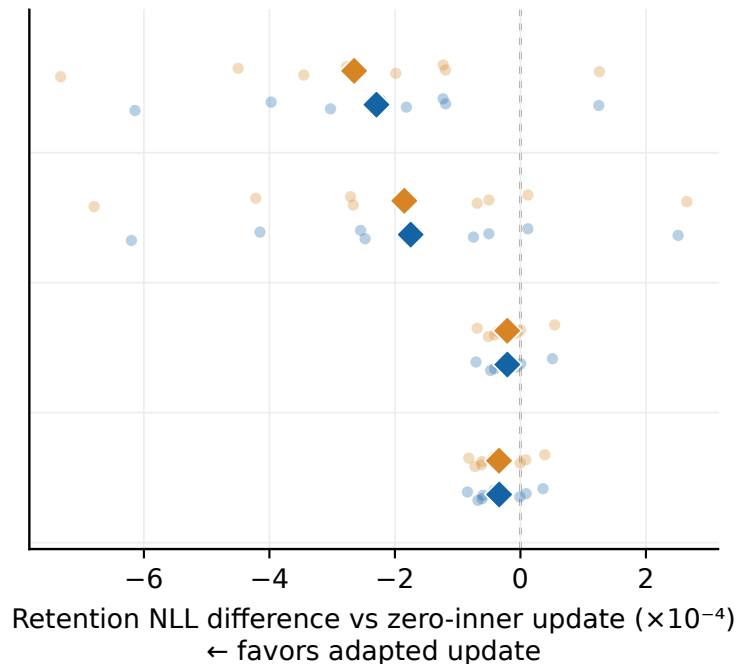
Does update direction explain the local retention effect?

◆ Adapted outer update ◆ Adapted direction, norm matched to zero-inner update

a Code



b Math



Fixed-quarter $\times 4$ · outer SGD 0.15 · transient inner rate $3e-5$

Dots: eight fixed training triples per checkpoint; diamonds: means. Local counterfactuals, one training seed.

Math retains 87–99% of its mean local benefit after norm matching. Code's terminal benefit is not explained here.